

**Post-Disaster Humanitarian Logistics and Casualty
Evacuation under Dynamic Network Disruptions: A Two-Stage
Distributionally Robust Optimization Approach**

By Li Ying, Hsiao

Department of Industrial Engineering and Engineering Management

National Tsing Hua University

Advisor: Tzu Li, Chen

06 / 2027

Abstract

Post-disaster humanitarian logistics operations are critically challenged by uncertain casualty demand and disrupted transportation networks following large-scale disasters. This study develops a two-stage moment-based distributionally robust optimization (DRO) model for the location and resource allocation of Casualty Collection Points (CCPs) under post-disaster uncertainty. In the first stage, strategic decisions are made regarding CCP establishment and the allocation of limited emergency personnel, ambulances, and relief supplies. In the second stage, these decisions are integrated with multi-period operational responses that capture dynamically evolving casualty flows with different injury severities across affected areas.

The proposed framework incorporates road network uncertainty by modeling damage-dependent network capacities and travel delays that affect ambulance routing and casualty evacuation over time. To address distributional ambiguity in disaster scenarios, a moment-based DRO approach is employed to hedge against worst-case realizations of casualty demand and post-disaster road condition uncertainty. Computational results demonstrate that the proposed approach can effectively reduce unmet evacuation demand, shorten treatment delays, and improve the robustness of post-disaster humanitarian response systems.

Table of Contents

Abstract	2
1. Introduction.....	5
1.1 Background and Motivation	5
1.2 Research Problem	5
1.3 Research Objectives.....	5
1.4 Research Contributions.....	5
1.5 Organization of the Thesis	5
2. Literature Review.....	5
2.1 Emergency Medical Service and Ambulance Dispatch	5
2.2 Disaster Relief Logistics and Road Damage	5
2.3 Robust Optimization and Distributionally Robust Optimization.....	5
2.4 Research Gaps.....	5
3. Problem Description	5
3.1 Problem Description	5
3.2 Assumptions.....	5
3.3 Sets, Parameters, and Decision Variables	6
3.4 Risk-averse two-stages stochastic programming.....	9
3.5 Distributionally robust optimization model.....	12
3.5.1 Ambiguity Sets.....	13
3.5.2 Model equivalent reformulations	14
5. Solution Methodology	16
5.1 Overall Solution Framework.....	16
5.2 Master Problem.....	16

5.3 Subproblem	16
5.4 Benders Decomposition or L-shaped Method	16
5.5 Algorithm Procedure	16
6. Numerical Experiments	16
6.1 Case Description and Data	16
6.2 Parameter Settings	16
6.3 Computational Results	16
6.4 Sensitivity Analysis	16
6.5 Out-of-Sample Performance	16
6.6 Managerial Insights	16
7. Conclusion	16
7.1 Summary of Findings	16
7.2 Theoretical and Practical Implications	16
7.3 Limitations and Future Research	16
References	16
Appendix A. Proofs	16
Appendix B. Additional Tables and Results	20

1. Introduction

1.1 Background and Motivation

1.2 Research Problem

1.3 Research Objectives

1.4 Research Contributions

1.5 Organization of the Thesis

2. Literature Review

2.1 Emergency Medical Service and Ambulance Dispatch

2.2 Disaster Relief Logistics and Road Damage

2.3 Robust Optimization and Distributionally Robust Optimization

2.4 Research Gaps

3. Problem Description

3.1 Problem Description

3.2 Assumptions

To formulate the post-disaster medical response and casualty evacuation problem, this study adopts the following modeling assumptions.

Assumption 1: Casualty classification and transportation requirements: Casualties are classified into three severity levels, namely minor, moderate, and severe, denoted by the set (L) . Minor casualties are assumed to be capable of reaching casualty collection points (CCPs) without ambulance assistance and do not require ambulance-based hospital transfer within the scope of this model. In contrast, moderate and severe casualties, represented by the subset $(L^{\wedge}\{Amb\} \subseteq L)$, require ambulance transportation from disaster areas to CCPs and subsequent transfer from CCPs to hospitals after receiving initial treatment or stabilization.

Assumption 2: Time- and scenario-dependent road availability: The post-disaster transportation network is subject to physical degradation and possible gradual recovery over

time. Road link availability is assumed to be both scenario-dependent and time-dependent. Rather than modeling each road link as either fully available or completely disrupted, this study represents the effective availability of each link by a continuous capacity proportion between 0 and 1. This formulation captures partial blockages, limited accessibility, and progressive road clearance operations after a disaster.

Assumption 3: Time- and scenario-dependent hospital receiving capacity: The receiving capacity of each hospital is assumed to vary across time periods and disaster scenarios. This reflects the operational uncertainty faced by hospitals after a disaster, including staff availability, medical resource depletion, facility damage, and gradual restoration of service capacity. Hospital receiving capacities are treated as exogenous uncertain parameters in the second-stage response problem.

Assumption 4: Within-period transportation approximation: Each discrete planning period is assumed to be sufficiently long relative to the travel time required on operational road links between disaster areas and CCPs, and between CCPs and hospitals. Therefore, casualty transportation and hospital transfer decisions are modeled as aggregate flows that can be completed within the same decision period. The effects of transportation difficulty are captured through road capacity availability and transportation cost or travel-time penalty parameters, without explicitly tracking en-route vehicles or inter-period travel states.

3.3 Sets, Parameters, and Decision Variables

Sets:

I	Set of disaster areas, indexed by i .
J	Set of candidate CCP locations, indexed by j .
H	Set of hospitals, indexed by h .
L	Set of casualty severity levels, indexed by l . Specifically, it includes minor, moderate, and severe levels.
L^{Amb}	Subset of casualty severity levels that require ambulance transportation and hospital transfer (i.e., moderate and severe casualties), $L^{Amb} \subseteq L$.
T	Set of discrete time periods, indexed by t .
S	Set of post-disaster scenarios, indexed by s .

First-Stage Deterministic Parameters:

f_j	Fixed cost of opening CCP j , $\forall j \in J$.
cv	Unit cost of assigning one medical staff member to a CCP.
ca	Unit cost of assigning one CCP ambulance to a CCP.

cy_{hj}	Unit cost of allocating one unit of medical supplies from hospital h to CCP j , $\forall j \in J, \forall h \in H$.
nv	Total number of available medical staff members.
na	Total number of available CCP ambulances.
$\bar{s}_h$	Maximum amount of medical supplies that hospital h can provide, $\forall h \in H$.
$\bar{v}_j$	Maximum number of medical staff members that can be assigned to CCP j , $\forall j \in J$.
$\bar{u}_j$	Maximum number of CCP ambulances that can be assigned to CCP j , $\forall j \in J$.
$\bar{y}_j$	Maximum amount of medical supplies that can be assigned to CCP j , $\forall j \in J$.

Second-Stage Deterministic Parameters:

c_{ij}	Normal maximum transportation capacity (casualties/period) on the road link from disaster area i to CCP j , $\forall i \in I, j \in J$.
c_{jh}	Normal maximum transportation capacity (casualties/period) on the road link from CCP j to hospital h , $\forall j \in J, h \in H$.
κ	Number of ambulance-requiring casualties that one CCP ambulance can transport in one period.
η	Number of casualties that one hospital ambulance can transfer in one period.
b_h	Number of ambulances owned by hospital h , $\forall h \in H$.
k_{jl}	Physical capacity of CCP j for casualties of severity level l , $\forall j \in J, l \in L$.
τ_l	Number of periods required to treat one casualty of severity level l at a CCP, $\forall l \in L$.
α_l	Number of casualties of severity level l that one medical staff member can treat in one period, $\forall l \in L$.
β_l	Amount of medical supplies consumed by one casualty of severity level l , $\forall l \in L$.
ρ_l	Unit penalty cost for one casualty of severity level l remaining in a disaster area for one period, $\forall l \in L$.
δ_l	Unit penalty cost for one treated casualty of severity level $\forall l \in L^{Amb}$ waiting at a CCP for hospital transfer for one period, $\forall l \in L^{Amb}$.
t_{ij}	Unit transportation cost, or cost-equivalent travel time penalty, for transporting one casualty from disaster area i to CCP j , $\forall i \in I, j \in J$.
t_{jh}	Unit transportation cost, or cost-equivalent travel time penalty, for transferring one casualty

from CCP j to hospital h , $\forall j \in J, h \in H$.

Random Variables:

- ξ_{ilt} Random variable representing the number of newly generated casualties of severity level l in disaster area i during period t . Its realization in scenario s is ξ_{ilts} .
- $\tilde{u}_{ijt}$ Random variable representing the available capacity proportion $([0,1])$ of the road link from disaster area i to CCP j during period t . Its realization in scenario s is u_{ijts} .
- $\tilde{w}_{jht}$ Random variable representing the available capacity proportion $([0,1])$ of the road link from CCP j to hospital h during period t . Its realization in scenario s is w_{jhst} .
- $\tilde{h}_{ht}$ Random variable representing the maximum number of casualties that hospital h can receive during period t . Its realization in scenario s is h_{hts} .

First-Stage Decision Variables:

- X_j $X_j \in \{0,1\}$, 1 if CCP j is opened, and 0 otherwise, $\forall j \in J$.
- V_j Number of medical staff members assigned to CCP j , $\forall j \in J$.
- U_j Number of CCP ambulances assigned to CCP j , $\forall j \in J$.
- Y_{hj} Amount of medical supplies allocated from hospital h to CCP j , $\forall j \in J, h \in H$.

Second-Stage Decision Variables:

- $FI_{ijt}(s)$ Number of casualties of severity level l transported from disaster area i to CCP j during period t under scenario s , $\forall i \in I, j \in J, l \in L, t \in T$.
- $FO_{jht}(s)$ Number of casualties of severity level $l \in L^{Amb}$ transferred from CCP j to hospital h during period t under scenario s , $\forall j \in J, h \in H, l \in L^{Amb}, t \in T$.
- $RM_{ilt}(s)$ Number of casualties of severity level l remaining in disaster area i at the end of period t under scenario s , $\forall i \in I, l \in L, t \in T$.
- $REG_{jlt}(s)$ Number of casualties of severity level l registered at CCP j during period t under scenario s , $\forall j \in J, l \in L, t \in T$.
- $TRT_{jlt}(s)$ Number of casualties of severity level l under treatment at CCP j during period t under scenario s , $\forall j \in J, l \in L, t \in T$.
- $WAT_{jlt}(s)$ Number of treated casualties of severity level $l \in L^{Amb}$ waiting at CCP j for hospital transfer at the end of period t under scenario s , $\forall j \in J, l \in L^{Amb}, t \in T$.

3.4 Risk-averse two-stages stochastic programming

The objective function is to minimize the first-stage preparation cost and the mean-CVaR of the second-stage recourse cost under a given scenario probability distribution:

$$\begin{aligned} \text{[RA-TSSP]} \quad \min \quad & \sum_{j \in J} f_j X_j + cv \sum_{j \in J} V_j + ca \sum_{j \in J} U_j + \sum_{h \in H} \sum_{j \in J} c y_{hj} Y_{hj} \\ & + \text{MCVaR}_\alpha(Q(X, V, U, Y; \tilde{\omega})) \end{aligned} \quad (1)$$

In (1), MCVaR represents a risk-averse evaluation of the uncertain post-disaster response cost under a given probability distribution. It is defined as follows:

$$\text{MCVaR}_\alpha(Q) = (1 - \lambda) \mathbb{E}_{\mathbf{p}}[Q] + \lambda \text{CVaR}_\alpha(Q), \quad 0 \leq \lambda \leq 1 \quad (1a)$$

The CVaR term is expressed in the following equivalent minimization form, where ϕ is the VaR threshold and $(a)^+ = \max\{a, 0\}$:

$$\text{CVaR}_\alpha(Q) = \min_{\phi \in \mathbb{R}} \left\{ \phi + \frac{1}{1 - \alpha} \mathbb{E}_{\mathbf{p}}[(Q - \phi)^+] \right\}, \quad 0 \leq \alpha < 1 \quad (1b)$$

The first-stage and second-stage constraints remain the same as in the original two-stage formulation and are listed in constraints (2)-(27).

The first-stage decisions are subject to the following constraints:

$$\sum_{j \in J} V_j \leq nv \quad (2)$$

$$\sum_{j \in J} U_j \leq na \quad (3)$$

$$\sum_{j \in J} Y_{hj} \leq \bar{s}_h, \quad \forall h \in H \quad (4)$$

$$V_j \leq \bar{v}_j X_j, \quad \forall j \in J \quad (5)$$

$$U_j \leq \bar{u}_j X_j, \quad \forall j \in J \quad (6)$$

$$\sum_{h \in H} Y_{hj} \leq \bar{y}_j X_j, \quad \forall j \in J \quad (7)$$

$$X_j \in \{0, 1\}, \quad \forall j \in J \quad (8)$$

$$V_j, U_j, Y_{hj} \in \mathbb{Z}_+, \quad \forall j \in J, h \in H \quad (9)$$

Constraint (2) ensures that the total number of medical staff members assigned to all opened CCPs does not exceed the available number of medical staff. Constraint (3) limits the total number of CCP ambulances assigned to all CCPs by the available CCP ambulance fleet.

Constraint (4) ensures that the amount of medical supplies allocated from each hospital does

not exceed its available supply. Constraint (5) allows medical staff to be assigned to CCP j only if CCP j is opened. Constraint (6) allows CCP ambulances to be assigned to CCP j only if CCP j is opened. Constraint (7) ensures that medical supplies can be allocated to CCP j only when CCP j is opened. Constraints (8) and (9) define the domains of the first-stage decision variables.

For each scenario s , the realization of uncertain casualty demand, road availability, and hospital receiving capacity is denoted by:

$$\omega^s = (\xi_{ilts}, u_{ijts}, w_{jhts}, h_{hts}), \quad \forall s \in S \quad (1c)$$

The scenario-dependent second-stage recourse cost is then written as:

$$Q_s = Q(X, V, U, Y; \omega^s), \quad \forall s \in S \quad (1d)$$

To linearize the positive-part term in CVaR, an auxiliary variable ℓ_s is introduced for each scenario:

$$\ell_s \geq Q_s - \phi, \quad \ell_s \geq 0, \quad \forall s \in S \quad (1e)$$

Given the first-stage decisions (X, V, U, Y) and the realization of scenario ω^s , the second-stage recourse function is defined as follows:

$$Q(X, V, U, Y; \omega_s) = \min \quad \sum_{t \in T} \sum_{i \in I} \sum_{l \in L} \rho_l RM_{ilt}(s) + \sum_{t \in T} \sum_{j \in J} \sum_{l \in L^{Amb}} \delta_l WAT_{jlt}(s) \quad (10)$$

$$+ \sum_{t \in T} \sum_{i \in I} \sum_{j \in J} \sum_{l \in L} t_{ij} FI_{ijlt}(s) + \sum_{t \in T} \sum_{j \in J} \sum_{h \in H} \sum_{l \in L^{Amb}} t_{jh} FO_{jhlt}(s)$$

$$\text{s.t.} \quad \sum_{l \in L} FI_{ijlt}(s) \leq c_{ij} u_{ijts} X_j, \quad \forall i \in I, j \in J, t \in T, s \in S \quad (11)$$

$$\sum_{l \in L^{Amb}} FO_{jhlt}(s) \leq c_{jh} w_{jhts} X_j, \quad \forall j \in J, h \in H, t \in T, s \in S \quad (12)$$

$$\sum_{i \in I} \sum_{l \in L^{Amb}} FI_{ijlt}(s) \leq \kappa U_j, \quad \forall j \in J, t \in T, s \in S \quad (13)$$

$$\sum_{j \in J} \sum_{l \in L^{Amb}} FO_{jhlt}(s) \leq \eta b_h, \quad \forall h \in H, t \in T, s \in S \quad (14)$$

$$RM_{ilt}(s) = RM_{il,t-1}(s) + \xi_{ilts} - \sum_{j \in J} FI_{ijlt}(s), \quad \forall i \in I, l \in L, t \in T, s \in S \quad (15)$$

$$RM_{il0}(s) = 0, \quad \forall i \in I, l \in L, s \in S \quad (16)$$

$$REG_{jlt}(s) = \sum_{i \in I} FI_{ijlt}(s), \quad \forall j \in J, l \in L, t \in T, s \in S \quad (17)$$

$$TRT_{jlt}(s) = \sum_{r=\max(1,t-\tau_l+1)}^t REG_{jlr}(s), \quad \forall j \in J, l \in L, t \in T, s \in S \quad (18)$$

$$WAT_{jlt}(s) = WAT_{jl,t-1}(s) + REG_{jl,t-\tau_l}(s) - \sum_{h \in H} FO_{jhlt}(s) \quad (19)$$

$$, j \in J, l \in L^{Amb}, t \in T, s \in S$$

$$WAT_{jl0}(s) = 0, \quad \forall j \in J, l \in L^{Amb}, s \in S \quad (20)$$

$$REG_{jl,t-\tau_l}(s) = 0 \quad \text{if } t - \tau_l < 1 \quad (21)$$

$$TRT_{jlt}(s) + WAT_{jlt}(s) \leq k_{jl}X_j, \quad \forall j \in J, l \in L^{Amb}, t \in T, s \in S \quad (22)$$

$$TRT_{jlt}(s) \leq k_{jl}X_j, \quad \forall j \in J, l \in L \setminus L^{Amb}, t \in T, s \in S \quad (23)$$

$$\sum_{l \in L} \frac{TRT_{jlt}(s)}{\alpha_l} \leq V_j, \quad \forall j \in J, t \in T, s \in S \quad (24)$$

$$\sum_{t \in T} \sum_{l \in L} \beta_l REG_{jlt}(s) \leq \sum_{h \in H} Y_{hj}, \quad \forall j \in J, s \in S \quad (25)$$

$$\sum_{j \in J} \sum_{l \in L^{Amb}} FO_{jhlt}(s) \leq h_{hts}, \quad \forall h \in H, t \in T, s \in S \quad (26)$$

$$FI_{ijlt}(s), FO_{jhlt}(s), RM_{ilt}(s), REG_{jlt}(s), TRT_{jlt}(s), WAT_{jlt}(s) \geq 0, \quad (27)$$

$$\forall i \in I, j \in J, h \in H, l \in L, t \in T, s \in S$$

The first term in (10) penalizes casualties remaining in disaster areas. The second term penalizes treated ambulance-requiring casualties waiting at CCPs for hospital transfer. The third term represents the transportation cost, or travel-time penalty, for moving casualties from disaster areas to CCPs. The fourth term represents the transportation cost, or travel-time penalty, for transferring ambulance-requiring casualties from CCPs to hospitals.

Constraints (11) and (12) capture the dynamic and partial availability of transportation links after a disaster. Constraint (13) limits the number of ambulance-requiring casualties transported from disaster areas to CCP j by the number of CCP ambulances assigned to CCP j . Constraint (14) limits the number of ambulance-requiring casualties transferred from CCPs to hospital h by the hospital ambulance fleet. Constraint (15) tracks the number of casualties remaining in each disaster area at the end of each period. Constraint (16) specifies the initial condition. Constraint (17) states that the number of casualties registered at CCP j equals the total number of casualties transported to CCP j from all disaster areas during the same period. Constraint (18) represents the number of casualties of severity level l that are under

treatment at CCP j during period t . A casualty remains under treatment for τ_l periods after registration. Constraint (19) tracks the number of treated ambulance-requiring casualties waiting at CCP j for hospital transfer. Constraint (20) defines the initial waiting condition, and constraint (21) specifies the boundary condition for treatment completion. Constraints (22)–(23) ensure that CCP capacity is respected at both the severity-specific level and the total facility level. Constraint (24) ensures that the treatment workload at each CCP does not exceed the treatment capacity provided by the assigned medical staff. Constraint (25) ensures that the total amount of medical supplies consumed by casualties registered at CCP j does not exceed the amount of supplies allocated to that CCP. Constraint (26) limits the number of casualties transferred to each hospital by the scenario-specific hospital receiving capacity in each period. Constraint (27) defines the non-negativity domains of the second-stage decision variables. For variables that are only defined for $l \in L^{Amb}$, such as $F_{jhl t}^{OUT}(s)$ and $Wait_{jlt}(s)$, the corresponding severity index is restricted to L^{Amb} .

3.5 Distributionally robust optimization model

The risk-averse two-stage stochastic programming model in Section 3.4 evaluates the recourse cost by MCVaR under a given probability distribution. However, in disaster response planning, the true scenario probability vector is usually difficult to estimate precisely. Therefore, this section extends the model to a distributionally robust optimization formulation by replacing MCVaR with the worst-case mean-CVaR over an ambiguity set of probability distributions.

$$\begin{aligned}
 \text{[RA-TSDRO]} \quad & \min \left[\sum_{j \in J} f_j X_j + cv \sum_{j \in J} V_j + ca \sum_{j \in J} U_j + \sum_{h \in H} \sum_{j \in J} c y_{hj} Y_{hj} \right. \\
 & \left. + \text{WMCVaR}_\alpha(Q(X, V, U, Y; \tilde{\omega})) \right] \\
 & \text{s.t. constraints (2) – (27)}
 \end{aligned} \tag{28}$$

For notational compactness, let $C^1(X, V, U, Y)$ denote the first-stage preparation cost, including the fixed CCP opening cost, medical staff assignment cost, CCP ambulance assignment cost, and medical supply allocation cost:

$$C^1(X, V, U, Y) = \sum_{j \in J} f_j X_j + cv \sum_{j \in J} V_j + ca \sum_{j \in J} U_j + \sum_{h \in H} \sum_{j \in J} c y_{hj} Y_{hj}$$

The WMCVaR term can be expanded using the mean-CVaR and CVaR definitions as follows:

$$\begin{aligned}
& \min \quad C^1(X, V, U, Y) + \text{WMCVaR}_\alpha(Q) \\
& = \min \quad C^1(X, V, U, Y) + \sup_{\mathbf{p} \in \mathcal{P}} \text{MCVaR}_\alpha(Q) \\
& = \min \quad C^1(X, V, U, Y) + \sup_{\mathbf{p} \in \mathcal{P}} \min_{\phi \in \mathbb{R}} \left\{ (1 - \lambda) \mathbb{E}_{\mathbf{p}}[Q] + \lambda \left[\phi + \frac{1}{1 - \alpha} \mathbb{E}_{\mathbf{p}}[(Q - \phi)^+] \right] \right\} \\
& \quad \text{s.t.} \quad \text{constraints (2) -- (27)}.
\end{aligned} \tag{29}$$

The objective function in the above formulation contains a $\min - \sup - \min$ structure, which is computationally intractable. Since the inner function is convex in ϕ and concave (linear) in $\mathbf{p}$, we can invoke **Sion's minimax theorem** to swap the operators $\sup_{\mathbf{p} \in \mathcal{P}}$ and $\min_{\phi \in \mathbb{R}}$. By doing so, the inner minimization over ϕ can be merged with the outer minimization of the first-stage decision variables.

Furthermore, to linearize the positive-part term $(Q - \phi)^+$, we introduce an auxiliary variable ℓ_s for each scenario $s \in S$. The model can then be equivalently reformulated as containing two worst-case expectation terms:

$$\mathbf{Q} = (Q_s)_{s \in S}, \quad \boldsymbol{\ell} = (\ell_s)_{s \in S}, \quad \mathbf{p} = (p_s)_{s \in S}, \quad \mathbb{E}_{\mathbf{p}}[Q] = \mathbf{Q}^\top \mathbf{p}, \quad \mathbb{E}_{\mathbf{p}}[(Q - \phi)^+] = \boldsymbol{\ell}^\top \mathbf{p}. \tag{30}$$

$$C^1 + \lambda \phi + (1 - \lambda) \sup_{\mathbf{p} \in \mathcal{P}} \mathbf{Q}^\top \mathbf{p} + \frac{\lambda}{1 - \alpha} \sup_{\mathbf{p} \in \mathcal{P}} \boldsymbol{\ell}^\top \mathbf{p} \tag{31}$$

3.5.1 Ambiguity Sets

This study considers three ambiguity sets for the finite scenario probability vector: box, ellipsoidal, and polyhedral ambiguity sets. In all definitions, $\mathbf{p}^0$ is the nominal probability vector estimated from available data, and $\mathbf{e}$ is a vector of ones.

Definition 1. Box ambiguity set.

$$\mathcal{P}^B = \{\mathbf{p} \mid \mathbf{p} = \mathbf{p}^0 + \boldsymbol{\epsilon}, \mathbf{e}^\top \boldsymbol{\epsilon} = 0, \|\boldsymbol{\epsilon}\|_\infty \leq \bar{\epsilon}_B\}. \tag{32}$$

The box ambiguity set controls the maximum absolute deviation of each scenario probability from its nominal value. The condition $\mathbf{e}^\top \boldsymbol{\epsilon} = 0$ ensures that the perturbed probabilities still sum to one.

Definition 2. Ellipsoidal ambiguity set.

$$\mathcal{P}^E = \{\mathbf{p} \mid \mathbf{p} = \mathbf{p}^0 + A_E \boldsymbol{\epsilon}, \mathbf{e}^\top A_E \boldsymbol{\epsilon} = 0, \mathbf{p}^0 + A_E \boldsymbol{\epsilon} \geq 0, \|\boldsymbol{\epsilon}\|_2 \leq 1\}. \quad (33)$$

The ellipsoidal ambiguity set controls the overall magnitude of the probability perturbation by using the Euclidean norm. The scaling matrix A_E determines the shape and size of the ambiguity region.

Definition 3. Polyhedral ambiguity set.

$$\mathcal{P}^P = \{\mathbf{p} \mid \mathbf{p} = \mathbf{p}^0 + A_P \boldsymbol{\epsilon}, \mathbf{e}^\top A_P \boldsymbol{\epsilon} = 0, \mathbf{p}^0 + A_P \boldsymbol{\epsilon} \geq 0, \|\boldsymbol{\epsilon}\|_1 \leq 1\}. \quad (34)$$

The polyhedral ambiguity set restricts the total absolute perturbation of the probability vector by using the 1-norm. This structure leads to a linear reformulation because the dual norm of the 1-norm is the infinity norm.

3.5.2 Model equivalent reformulations

The RA-TSDRO model in (28)-(31) contains worst-case expectation terms over the ambiguity set and is therefore reformulated into tractable deterministic equivalents. Detailed derivations are provided in Appendix A.

Theorem 1. If the probability vector belongs to the box ambiguity set, the model can be reformulated as the following mixed-integer linear programming model:

$$\begin{aligned} [\text{DRO-B}] \min \quad & C^1 + \lambda \phi + (1 - \lambda)[\mathbf{Q}^\top \mathbf{p}^0 + \bar{\epsilon}_B \mathbf{e}^\top \boldsymbol{\zeta} + \bar{\epsilon}_B \mathbf{e}^\top \boldsymbol{\omega}] \\ & + \frac{\lambda}{1 - \alpha} [\boldsymbol{\ell}^\top \mathbf{p}^0 + \bar{\epsilon}_B \mathbf{e}^\top \hat{\boldsymbol{\zeta}} + \bar{\epsilon}_B \mathbf{e}^\top \hat{\boldsymbol{\omega}}] \\ \text{s.t.} \quad & e\gamma + \boldsymbol{\zeta} - \boldsymbol{\omega} = \mathbf{Q}, \quad e\hat{\gamma} + \hat{\boldsymbol{\zeta}} - \hat{\boldsymbol{\omega}} = \boldsymbol{\ell}, \\ & \ell_s \geq Q_s - \phi, \quad \ell_s \geq 0, \quad \forall s \in S, \\ & \boldsymbol{\zeta}, \boldsymbol{\omega}, \hat{\boldsymbol{\zeta}}, \hat{\boldsymbol{\omega}} \geq 0, \\ & \text{constraints (2) - (27)}. \end{aligned} \quad (35)$$

Theorem 2. If the probability vector belongs to the ellipsoidal ambiguity set, the model can be reformulated as the following second-order cone programming model:

$$\begin{aligned}
[\text{DRO-E}] \min \quad & C^1 + \lambda\phi + (1 - \lambda)[\mathbf{Q}^\top \mathbf{p}^0 + (\mathbf{p}^0)^\top \Delta + \nu] \\
& + \frac{\lambda}{1 - \alpha} [\boldsymbol{\ell}^\top \mathbf{p}^0 + (\mathbf{p}^0)^\top \hat{\Delta} + \hat{\nu}] \\
\text{s.t.} \quad & \|A_E^\top \mathbf{Q} + A_E^\top \Delta + \pi A_E^\top e\|_2 \leq \nu, \\
& \|A_E^\top \boldsymbol{\ell} + A_E^\top \hat{\Delta} + \hat{\pi} A_E^\top e\|_2 \leq \hat{\nu}, \\
& \ell_s \geq Q_s - \phi, \quad \ell_s \geq 0, \quad \forall s \in S, \\
& \Delta, \hat{\Delta} \geq 0, \quad \nu, \hat{\nu} \geq 0, \\
& \text{constraints (2) -- (27)}.
\end{aligned} \tag{36}$$

Theorem 3. If the probability vector belongs to the polyhedral ambiguity set, the model can be reformulated as the following mixed-integer linear programming model:

$$\begin{aligned}
[\text{DRO-P}] \min \quad & C^1 + \lambda\phi + (1 - \lambda)[\mathbf{Q}^\top \mathbf{p}^0 + (\mathbf{p}^0)^\top \Gamma + \psi] \\
& + \frac{\lambda}{1 - \alpha} [\boldsymbol{\ell}^\top \mathbf{p}^0 + (\mathbf{p}^0)^\top \hat{\Gamma} + \hat{\psi}] \\
\text{s.t.} \quad & \|A_P^\top \mathbf{Q} + A_P^\top \Gamma + \chi A_P^\top e\|_\infty \leq \psi, \\
& \|A_P^\top \boldsymbol{\ell} + A_P^\top \hat{\Gamma} + \hat{\chi} A_P^\top e\|_\infty \leq \hat{\psi}, \\
& \ell_s \geq Q_s - \phi, \quad \ell_s \geq 0, \quad \forall s \in S, \\
& \Gamma, \hat{\Gamma} \geq 0, \quad \psi, \hat{\psi} \geq 0, \\
& \text{constraints (2) -- (27)}.
\end{aligned} \tag{37}$$

The infinity-norm constraints in the polyhedral reformulation can be equivalently expressed as pairs of linear inequalities. Therefore, the polyhedral formulation is computationally tractable using standard mixed-integer linear programming solvers.

5. Solution Methodology

5.1 Overall Solution Framework

5.2 Master Problem

5.3 Subproblem

5.4 Benders Decomposition or L-shaped Method

5.5 Algorithm Procedure

6. Numerical Experiments

6.1 Case Description and Data

6.2 Parameter Settings

6.3 Computational Results

6.4 Sensitivity Analysis

6.5 Out-of-Sample Performance

6.6 Managerial Insights

7. Conclusion

7.1 Summary of Findings

7.2 Theoretical and Practical Implications

7.3 Limitations and Future Research

References

Appendix A. Proofs

This appendix provides the deterministic equivalent reformulations of the DRO models presented in Section 3.5.2. The proof follows the standard structure used in risk-averse two-stage DRO reformulations: first, the WMCVaR term is decomposed into two worst-case expectation problems; then each inner maximization problem is reformulated by linear programming duality or Lagrange duality, depending on the ambiguity set.

$$\begin{aligned}
\text{WMCVaR}_\alpha(\mathbf{Q}) &= \sup_{\mathbf{p} \in \mathcal{P}} \min_{\phi \in \mathbb{R}} \left\{ (1 - \lambda) \mathbb{E}_{\mathbf{p}}[Q] + \lambda \left[\phi + \frac{1}{1 - \alpha} \mathbb{E}_{\mathbf{p}}[(Q - \phi)^+] \right] \right\} \\
&= \min_{\phi \in \mathbb{R}} \left\{ \lambda \phi + (1 - \lambda) \sup_{\mathbf{p} \in \mathcal{P}} \mathbf{Q}^\top \mathbf{p} + \frac{\lambda}{1 - \alpha} \sup_{\mathbf{p} \in \mathcal{P}} \boldsymbol{\ell}^\top \mathbf{p} \right\}
\end{aligned} \tag{A.1}$$

Therefore, for each ambiguity set, it is sufficient to reformulate the following two inner maximization problems: the worst-case expected recourse cost and the worst-case expected CVaR positive-part term.

Theorem A.1 (DRO-B).

If the probability vector belongs to the box ambiguity set, the RA-TSDRO model can be equivalently reformulated as the following mixed-integer linear programming model:

$$\begin{aligned}
[\text{DRO-B}] \min \quad & C^1 + \lambda \phi + (1 - \lambda) [\mathbf{Q}^\top \mathbf{p}^0 + \bar{\epsilon}_B \mathbf{e}^\top \boldsymbol{\zeta} + \bar{\epsilon}_B \mathbf{e}^\top \boldsymbol{\varpi}] \\
& + \frac{\lambda}{1 - \alpha} [\boldsymbol{\ell}^\top \mathbf{p}^0 + \bar{\epsilon}_B \mathbf{e}^\top \hat{\boldsymbol{\zeta}} + \bar{\epsilon}_B \mathbf{e}^\top \hat{\boldsymbol{\varpi}}] \\
\text{s.t.} \quad & e\gamma + \boldsymbol{\zeta} - \boldsymbol{\varpi} = \mathbf{Q}, \quad e\hat{\gamma} + \hat{\boldsymbol{\zeta}} - \hat{\boldsymbol{\varpi}} = \boldsymbol{\ell}, \\
& \ell_s \geq Q_s - \phi, \quad \ell_s \geq 0, \quad \forall s \in S, \\
& \boldsymbol{\zeta}, \boldsymbol{\varpi}, \hat{\boldsymbol{\zeta}}, \hat{\boldsymbol{\varpi}} \geq 0, \\
& \text{constraints (2) -- (27)}.
\end{aligned} \tag{A.2}$$

Proof.

Under the box ambiguity set, the probability vector is expressed as $\mathbf{p} = \mathbf{p}^0 + \epsilon \mathbf{e}$. Starting from the first inner maximization problem, we obtain:

$$\sup_{\mathbf{p} \in \mathcal{P}^B} \mathbf{Q}^\top \mathbf{p} = \mathbf{Q}^\top \mathbf{p}^0 + \max_{\boldsymbol{\epsilon}} \{ \mathbf{Q}^\top \boldsymbol{\epsilon} \mid \mathbf{e}^\top \boldsymbol{\epsilon} = 0, -\bar{\epsilon}_B \mathbf{e} \leq \boldsymbol{\epsilon} \leq \bar{\epsilon}_B \mathbf{e} \}. \tag{A.3}$$

The last maximization problem is a linear program with respect to the perturbation vector ϵ . By linear programming duality, the equivalent dual formulation is:

$$\begin{aligned}
\min_{\gamma, \boldsymbol{\zeta}, \boldsymbol{\varpi}} \quad & \mathbf{Q}^\top \mathbf{p}^0 + \bar{\epsilon}_B \mathbf{e}^\top \boldsymbol{\zeta} + \bar{\epsilon}_B \mathbf{e}^\top \boldsymbol{\varpi} \\
\text{s.t.} \quad & e\gamma + \boldsymbol{\zeta} - \boldsymbol{\varpi} = \mathbf{Q}, \\
& \boldsymbol{\zeta}, \boldsymbol{\varpi} \geq 0.
\end{aligned} \tag{A.4}$$

The second inner maximization problem has the same structure. Replacing $\mathbf{Q}$ by $\boldsymbol{\ell}$ yields:

$$\begin{aligned}
\sup_{\mathbf{p} \in \mathcal{P}^B} \boldsymbol{\ell}^\top \mathbf{p} &= \min_{\hat{\gamma}, \hat{\zeta}, \hat{\omega}} \boldsymbol{\ell}^\top \mathbf{p}^0 + \bar{\epsilon}_B \mathbf{e}^\top \hat{\zeta} + \bar{\epsilon}_B \mathbf{e}^\top \hat{\omega} \\
\text{s.t. } &\mathbf{e} \hat{\gamma} + \hat{\zeta} - \hat{\omega} = \boldsymbol{\ell}, \\
&\hat{\zeta} \geq \mathbf{0}, \quad \hat{\omega} \geq \mathbf{0}.
\end{aligned} \tag{A.5}$$

Substituting these two dual formulations into (A.1) gives the DRO-B deterministic equivalent model. This completes the proof of Theorem A.1.

Theorem A.2 (DRO-E).

If the probability vector belongs to the ellipsoidal ambiguity set, the RA-TSDRO model can be equivalently reformulated as the following second-order cone programming model:

$$\begin{aligned}
[\text{DRO-E}] \min \quad & C^1 + \lambda \phi + (1 - \lambda)[\mathbf{Q}^\top \mathbf{p}^0 + (\mathbf{p}^0)^\top \Delta + \nu] \\
& + \frac{\lambda}{1 - \alpha} [\boldsymbol{\ell}^\top \mathbf{p}^0 + (\mathbf{p}^0)^\top \hat{\Delta} + \hat{\nu}] \\
\text{s.t. } \quad & \|A_E^\top \mathbf{Q} + A_E^\top \Delta + \pi A_E^\top \mathbf{e}\|_2 \leq \nu, \\
& \|A_E^\top \boldsymbol{\ell} + A_E^\top \hat{\Delta} + \hat{\pi} A_E^\top \mathbf{e}\|_2 \leq \hat{\nu}, \\
& \ell_s \geq Q_s - \phi, \quad \ell_s \geq 0, \quad \forall s \in S, \\
& \Delta, \hat{\Delta} \geq 0, \quad \nu, \hat{\nu} \geq 0, \\
& \text{constraints (2) -- (27)}.
\end{aligned} \tag{A.6}$$

Proof.

Under the ellipsoidal ambiguity set, the probability vector is expressed as $\mathbf{p} = \mathbf{p}^0 + \mathbf{A}_E \boldsymbol{\epsilon}$. Starting from the first inner maximization problem, we obtain:

$$\sup_{\mathbf{p} \in \mathcal{P}^E} \mathbf{Q}^\top \mathbf{p} = \mathbf{Q}^\top \mathbf{p}^0 + \max_{\boldsymbol{\epsilon}} \{ \mathbf{Q}^\top \mathbf{A}_E \boldsymbol{\epsilon} \mid \mathbf{e}^\top \mathbf{A}_E \boldsymbol{\epsilon} = 0, \mathbf{p}^0 + \mathbf{A}_E \boldsymbol{\epsilon} \geq 0, \|\boldsymbol{\epsilon}\|_2 \leq 1 \}. \tag{A.7}$$

Introduce the Lagrange multipliers $\Delta \geq 0$ for the probability non-negativity constraint, $\nu \geq 0$ for the ellipsoidal norm constraint, and π as a free multiplier for the probability-sum constraint. The Lagrangian is:

$$\begin{aligned}
\mathcal{L}(\boldsymbol{\epsilon}, \Delta, \nu, \pi) &= \mathbf{Q}^\top \mathbf{A}_E \boldsymbol{\epsilon} + \Delta^\top (\mathbf{p}^0 + \mathbf{A}_E \boldsymbol{\epsilon}) + \pi \mathbf{e}^\top \mathbf{A}_E \boldsymbol{\epsilon} + \nu(1 - \|\boldsymbol{\epsilon}\|_2) \\
&= (\mathbf{p}^0)^\top \Delta + \nu + (\mathbf{A}_E^\top \mathbf{Q} + \mathbf{A}_E^\top \Delta + \pi \mathbf{A}_E^\top \mathbf{e})^\top \boldsymbol{\epsilon} - \nu \|\boldsymbol{\epsilon}\|_2
\end{aligned} \tag{A.8}$$

The inner maximization over $\boldsymbol{\epsilon}$ is finite only when the linear term is dominated by the

Euclidean-norm penalty. Equivalently, using the dual norm relationship of the Euclidean norm:

$$\max_{\boldsymbol{\varepsilon}} \{\mathbf{b}^\top \boldsymbol{\varepsilon} - \nu \|\boldsymbol{\varepsilon}\|_2\} = \begin{cases} 0, & \|\mathbf{b}\|_2 \leq \nu, \\ +\infty, & \text{otherwise,} \end{cases} \text{ where } \mathbf{b} = A_E^\top \mathbf{Q} + A_E^\top \boldsymbol{\Delta} + \pi A_E^\top \mathbf{e}. \quad (\text{A.9})$$

Thus, the first worst-case expectation term and the second positive-part term have the following dual representations:

$$\begin{aligned} \sup_{\mathbf{p} \in \mathcal{P}^E} \mathbf{Q}^\top \mathbf{p} &= \min_{\boldsymbol{\Delta}, \nu, \pi} \{ \mathbf{Q}^\top \mathbf{p}^0 + (\mathbf{p}^0)^\top \boldsymbol{\Delta} + \nu \mid \|\mathbf{A}_E^\top \mathbf{Q} + \mathbf{A}_E^\top \boldsymbol{\Delta} + \pi \mathbf{A}_E^\top \mathbf{e}\|_2 \leq \nu, \boldsymbol{\Delta} \geq \mathbf{0}, \nu \geq 0 \} \quad (\text{A.10}) \\ \sup_{\mathbf{p} \in \mathcal{P}^E} \boldsymbol{\ell}^\top \mathbf{p} &= \min_{\hat{\boldsymbol{\Delta}}, \hat{\nu}, \hat{\pi}} \{ \boldsymbol{\ell}^\top \mathbf{p}^0 + (\mathbf{p}^0)^\top \hat{\boldsymbol{\Delta}} + \hat{\nu} \mid \|\mathbf{A}_E^\top \boldsymbol{\ell} + \mathbf{A}_E^\top \hat{\boldsymbol{\Delta}} + \hat{\pi} \mathbf{A}_E^\top \mathbf{e}\|_2 \leq \hat{\nu}, \hat{\boldsymbol{\Delta}} \geq \mathbf{0}, \hat{\nu} \geq 0 \}. \end{aligned}$$

Substituting these two dual formulations into (A.1) gives the DRO-E deterministic equivalent model. This completes the proof of Theorem A.2.

Theorem A.3 (DRO-P).

If the probability vector belongs to the polyhedral ambiguity set, the RA-TSDRO model can be equivalently reformulated as the following mixed-integer linear programming model:

$$\begin{aligned} [\text{DRO-P}] \quad \min \quad & C^1 + \lambda \phi + (1 - \lambda) [\mathbf{Q}^\top \mathbf{p}^0 + (\mathbf{p}^0)^\top \Gamma + \psi] \\ & + \frac{\lambda}{1 - \alpha} [\boldsymbol{\ell}^\top \mathbf{p}^0 + (\mathbf{p}^0)^\top \hat{\Gamma} + \hat{\psi}] \\ \text{s.t.} \quad & \|\mathbf{A}_P^\top \mathbf{Q} + \mathbf{A}_P^\top \Gamma + \chi \mathbf{A}_P^\top \mathbf{e}\|_\infty \leq \psi, \\ & \|\mathbf{A}_P^\top \boldsymbol{\ell} + \mathbf{A}_P^\top \hat{\Gamma} + \hat{\chi} \mathbf{A}_P^\top \mathbf{e}\|_\infty \leq \hat{\psi}, \\ & \ell_s \geq Q_s - \phi, \quad \ell_s \geq 0, \quad \forall s \in S, \\ & \Gamma, \hat{\Gamma} \geq 0, \quad \psi, \hat{\psi} \geq 0, \\ & \text{constraints (2) -- (27)}. \end{aligned} \quad (\text{A.11})$$

Proof.

Under the polyhedral ambiguity set, the probability vector is expressed as $\mathbf{p} = \mathbf{p}^0 + \mathbf{A}_P \boldsymbol{\varepsilon}$. Starting from the first inner maximization problem, we obtain:

$$\sup_{\mathbf{p} \in \mathcal{P}^P} \mathbf{Q}^\top \mathbf{p} = \mathbf{Q}^\top \mathbf{p}^0 + \max_{\boldsymbol{\varepsilon}} \{ \mathbf{Q}^\top \mathbf{A}_P \boldsymbol{\varepsilon} \mid \mathbf{e}^\top \mathbf{A}_P \boldsymbol{\varepsilon} = 0, \mathbf{p}^0 + \mathbf{A}_P \boldsymbol{\varepsilon} \geq 0, \|\boldsymbol{\varepsilon}\|_1 \leq 1 \}. \quad (\text{A.12})$$

Introduce the Lagrange multipliers $\Gamma \geq 0$ for the probability non-negativity constraint, $\psi \geq 0$ for the 1-norm constraint, and χ as a free multiplier for the probability-sum constraint. The Lagrangian is:

$$\begin{aligned}
\mathcal{L}(\boldsymbol{\epsilon}, \boldsymbol{\Gamma}, \psi, \chi) &= \mathbf{Q}^\top \mathbf{A}_P \boldsymbol{\epsilon} + \boldsymbol{\Gamma}^\top (\mathbf{p}^0 + \mathbf{A}_P \boldsymbol{\epsilon}) + \chi \mathbf{e}^\top \mathbf{A}_P \boldsymbol{\epsilon} + \psi(1 - \|\boldsymbol{\epsilon}\|_1) \\
&= (\mathbf{p}^0)^\top \boldsymbol{\Gamma} + \psi + (\mathbf{A}_P^\top \mathbf{Q} + \mathbf{A}_P^\top \boldsymbol{\Gamma} + \chi \mathbf{A}_P^\top \mathbf{e})^\top \boldsymbol{\epsilon} - \psi \|\boldsymbol{\epsilon}\|_1.
\end{aligned} \tag{A.13}$$

The conjugate of the 1-norm is the infinity norm. Hence, the inner maximization over epsilon is finite only under the following condition:

$$\max_{\boldsymbol{\epsilon}} \{\mathbf{b}^\top \boldsymbol{\epsilon} - \psi \|\boldsymbol{\epsilon}\|_1\} = \begin{cases} 0, & \|\mathbf{b}\|_\infty \leq \psi, \\ +\infty, & \text{otherwise,} \end{cases} \text{ where } \mathbf{b} = \mathbf{A}_P^\top \mathbf{Q} + \mathbf{A}_P^\top \boldsymbol{\Gamma} + \chi \mathbf{A}_P^\top \mathbf{e}. \tag{A.14}$$

Therefore, the first worst-case expectation term and the second positive-part term can be written as:

$$\begin{aligned}
& \sup_{\mathbf{p} \in \mathcal{P}^P} \mathbf{Q}^\top \mathbf{p} \\
&= \min_{\boldsymbol{\Gamma}, \psi, \chi} \{ \mathbf{Q}^\top \mathbf{p}^0 + (\mathbf{p}^0)^\top \boldsymbol{\Gamma} + \psi \mid \|\mathbf{A}_P^\top \mathbf{Q} + \mathbf{A}_P^\top \boldsymbol{\Gamma} + \chi \mathbf{A}_P^\top \mathbf{e}\|_\infty \leq \psi, \boldsymbol{\Gamma} \geq \mathbf{0}, \psi \geq 0 \}, \\
& \sup_{\mathbf{p} \in \mathcal{P}^P} \boldsymbol{\ell}^\top \mathbf{p} \\
&= \min_{\hat{\boldsymbol{\Gamma}}, \hat{\psi}, \hat{\chi}} \{ \boldsymbol{\ell}^\top \mathbf{p}^0 + (\mathbf{p}^0)^\top \hat{\boldsymbol{\Gamma}} + \hat{\psi} \mid \|\mathbf{A}_P^\top \boldsymbol{\ell} + \mathbf{A}_P^\top \hat{\boldsymbol{\Gamma}} + \hat{\chi} \mathbf{A}_P^\top \mathbf{e}\|_\infty \leq \hat{\psi}, \hat{\boldsymbol{\Gamma}} \geq \mathbf{0}, \hat{\psi} \geq 0 \}.
\end{aligned} \tag{A.15}$$

The infinity-norm constraints can be implemented as linear inequalities:

$$\|\mathbf{b}\|_\infty \leq \psi \Leftrightarrow -\psi \leq b_s \leq \psi, \quad \forall s \in S. \tag{34}$$

Substituting these two dual formulations into (A.1) gives the DRO-P deterministic equivalent model. This completes the proof of Theorem A.3.

Appendix B. Additional Tables and Results